HTTP Errors DevOps Engineers Should Troubleshoot

🔥 5xx – Server Errors (High Priority for DevOps)

These almost always mean something is wrong with **your infrastructure or services**.

Code	What It Indicates	DevOps Focus Area
500	Internal Server Error	App crashed, unhandled exception. Check logs, backend service.
502	Bad Gateway	Upstream server (e.g., app behind NGINX, API gateway) is down or returning errors.
503	Service Unavailable	Server overloaded or in maintenance. Watch scaling, health checks, availability zones.
504	Gateway Timeout	Timeout between gateway and backend. Check backend performance, timeouts, or DNS.
501	Not Implemented (less common)	Feature not deployed or misconfigured routing.

⚠️ 4xx – Client Errors (Important in Context)

These are usually **application-side or user-side**, but some are **important to DevOps** for **monitoring abuse, rate-limiting, or misconfiguration**.

PATHNEX
Shaping Careers, Building Futures

Code	What It Indicates	DevOps Focus Area
400	Bad Request	Often caused by malformed requests. If frequent, may be due to a proxy, WAF, or misrouted request.
401/403	Unauthorized/Forbidden	Auth misconfigurations, missing headers, IAM or firewall rules.
404	Not Found	Might point to broken routing, DNS, or misdeployment.
408	Request Timeout	Could point to app latency or network issues.
429	Too Many Requests	Rate limiting triggered — check for DoS, misconfigured throttling, or autoscaling thresholds.

🛠️ DevOps Action Plan

Error	Tools/Areas to Check
500	Logs (e.g., ELK, Loki), APM (Datadog, New Relic), app container status
502	NGINX/HAProxy logs, service mesh (e.g., Istio), upstream health
503	Load balancer health checks, auto-scaling, resource limits (CPU/RAM)
504	Backend timeouts, DNS resolution, inter-service latency
429	Rate limits in API Gateway, throttling policies, bot traffic detection
408	Application responsiveness, timeout settings, network latency